Enhancing Vehicle Speed Estimation Accuracy Using Computer Vision

Oluwatomisin Isogun and Modupe Agagu
Department of Computer Science, Olusegun Agagu University of Science & Technology, Nigeria

Abstract—**In today's transportation world, accurately determining vehicle speeds is crucial for effective traffic management and enhancing road safety. However, achieving precise speed measurements poses significant challenges, especially when using wide-angle cameras prone to perspective distortion, where distant objects appear disproportionately smaller than those closer to the camera. To address this issue, our approach integrates core computer vision techniques, including object detection, object tracking, polygon zone mapping, and perspective correction using OpenCV, to estimate the speeds of tracked vehicles accurately. The model was rigorously evaluated on several performance metrics such as accuracy, precision, recall and F1 score. It achieved an accuracy of 0.80, correctly predicting 80% of vehicle speeds. The precision of the model was high, at 0.90, indicating that 90% of the predicted correct speed estimations were indeed accurate. The model also demonstrated a strong recall of 0.82, effectively identifying 82% of the true positives for correct speed estimations. Additionally, the F1 Score, which balances precision and recall, was 0.86, underscoring the model's capability to minimize both false positives and false negatives while maintaining overall effectiveness. This step-by-step analysis and implementation significantly improved speed estimation accuracy, particularly in scenarios affected by perspective distortion, making it a robust solution for real-world traffic monitoring systems.**

Keywords: Computer Vision, Vehicle Speed Estimation, Perspective Distortion, Machine Learning

1. INTRODUCTION

Over the years, vehicle speed estimation systems have played a crucial role in modern society, particularly in urban environments, where traffic management and safety are paramount concerns. These systems offer numerous benefits, including enhanced traffic flow, reduced congestion, improved road safety, and more efficient law enforcement. By accurately measuring vehicle speeds, authorities can enforce speed limits effectively, detect traffic violations, and optimize traffic signal timing to minimize delays and improve overall transportation efficiency. However, one significant challenge faced by existing vehicle speed estimation systems is perspective distortion, particularly when utilizing wide-angle cameras for data capture (Hua *et al.,* 2018). Perspective distortion occurs when objects appear differently in size or shape than they actually are, depending on their distance from the camera and the angle of view. In the context of vehicle speed estimation, perspective distortion can significantly affect the accuracy of speed measurements, leading to errors in the recorded data. In simple terms, perspective distortion can be understood as the phenomenon where objects closer to the camera appear larger than those farther away, even though they may be similar in size in reality. This distortion becomes increasingly associated with wide-angle cameras, which are commonly used in traffic surveillance systems to capture a broader field of view (Zhang *et al.,* 2020). Given the detrimental impact of perspective distortion on the accuracy of vehicle speed estimation systems, the focus of this project is to develop an algorithm capable of mitigating this distortion and thereby enhancing the accuracy of speed measurements.

The motivation behind this research stems from the increasing need for advanced safety measures and efficient traffic management systems in our rapidly urbanizing societies. Vehicle speed estimation plays a pivotal role in the realm of traffic surveillance (Ashraf *et al.,* 2023), accident analysis, and law enforcement, providing a technological backbone for initiatives aimed at enhancing road safety and reducing traffic-related fatalities (Gunawan *et al.,* 2019).

With the global surge in vehicle numbers and urban traffic congestion, the demand for accurate and reliable vehicle speed estimation systems has never been more critical. However, the accuracy of these systems is frequently compromised by perspective distortion, particularly when utilizing wide-angle cameras that are commonly deployed in traffic monitoring setups. Perspective distortion, a phenomenon where objects appear smaller or larger depending on their distance from the lens, can significantly skew speed measurements, leading to unreliable data and potentially undermining the effectiveness of speed enforcement campaigns and traffic management strategies. The importance of addressing this issue lies in the potential of accurate speed estimation contributing to safer urban environments, ensuring that speed limits are effectively monitored and enforced, deter speeding, reduce the occurrence of accidents, and save lives. Furthermore, accurate speed data is invaluable for urban planners and traffic engineers, facilitating the design of safer roads and more efficient traffic flow patterns. This research is thus motivated by the pressing need for an innovative solution that can overcome the challenges posed by perspective distortion, enhancing the accuracy and reliability of vehicle speed estimation systems. By developing an algorithm that mitigates the effects of this distortion, this study will contribute significantly to the advancement of traffic monitoring technologies, offering promising implications for road safety, law enforcement, and urban traffic management.

2. RELATED WORK

Tracking vehicle speeds by object detection is widely acknowledged in existing vehicle tracking studies, consequently, a review of vehicle detection and tracking conducted covers the overview of vehicle estimation systems, and challenges in perspective distortion.

A. Overview of Vehicle Speed Estimation Systems

Due to the hard work of many researchers and scientists, we now have various methods of determining moving vehicle speed. They are Inductive loops, RADAR gun, LASER Gun, LiDAR, Manual count (Yu *et al.,* 2022). But these intrusive and non-intrusive sensors are either too expensive and requires high maintenance or damages the already existing road. So, to overcome the limitations in existing methods, several video-based approaches were proposed for estimating or measuring the speed of vehicles in roadways. Almost all the methods include a background or foreground segmentation step to detect image regions containing motion. The methods of background extraction from color image (Afifah *et al.,* 2019) based on average value, median filter (Bell *et al.,* 2020) and common region. Few other common approaches for this task include simple frame differences, as well as statistic models based on Gaussian distributions (Najman *et al.,* 2020) or other measures. Image tracking is one of the steps to estimate vehicle speed. Some of the common tracking method involves image patches (Kumar *et al.,* 2018), edges (Wen *et al.,* 2015), blobs (Shi *et al.,* 2018) and corners. Few researchers have successfully tracked vehicles based on the license plate region, or a combination of such features. Some methods are sensitive to conditions such as shadows, perspective distortion and illumination variation. Blob analysis is one such method which can have high tracking error. Moreover, these methods produce satisfactory results only when the camera is positioned high above the roadway, with the blobs being tracked for many frames. Many researchers have found ways to avoid the problem associated with blob analysis. Huang *et al.,* (2021), in their work published in the *2021 International Conference on 3D Vision (3DV)* stresses the key idea of incorporating motion prior knowledge into simultaneous optimization of extrinsic camera parameters and also the introduction of physics-geometry consistency to reduce the low and high frequency noises of the detected object semantics. Experimental results show that accurate camera parameters and object motions can be obtained through one-stage optimization. However, the disadvantage of using this algorithm is that it can handle only a single vehicle at

a time. Moreover, they require a side view of the vehicles and do not take perspective into account. Vehicle detection and tracking is one of the most important steps to estimate vehicle speed. While using different methods and algorithms, vehicles might go undetected or wrongfully detected. To remove misdetection of vehicle (Yang *et al.,* 2019) due to vehicle travelling from other lane, or other small movements such as tree waving can be avoided using ROI (regions of interest) extraction. The background is multiplied with ROI mask. So that vehicles are detected accurately. Furthermore, thresholding and morphological operations are used to reduce noise. In thresholding selection of threshold value is based on various methods. The threshold value can be selected manually or automatically by using thresholding.

B. Traditional Approaches to Vehicle Speed Estimation

In Bell et al. 2020, a workflow was created by which vehicle speeds are estimated semi-automatically via a fixed DSLR camera. Deep learning algorithms such as YOLOv2 was used for vehicle detection, while "Simple Online Realtime Tracking" (SORT) algorithm enabled for tracking of vehicles. The authors believed that perspective projection and scaling was a factor that needs to be dealt with in order for vehicle speed estimation systems to have an increased accuracy and ensure reliability. To achieve that, an experimental framework for analytics was developed, where deep learning, tracking, and image mapping systems can undergo optimization for fast and seamless operation in a combined processing workflow. Although traditional computer vision techniques often use "Haar-like" classifiers and "AdaBoost" machine learning algorithms for vehicle detection such as the ones reviewed by Wen *et al.* (2015). However, more recently, with the use of deep learning, Tang *et al.* (2018), achieved flow characterization (including speed estimation), and object re-identification. The vehicle detection system utilized YOLOv2 deep learning algorithm to recognize vehicles across various categories which ranges from creating 2D bounding boxes, foot print coordinates (midpoint of lower corners) to the transformation into 3D space preceding speed estimation. A common approach used for correcting perspective distortion effects - that results from camera position and orientation relative to the road surface - is by manually labelling two pairs of parallel lines, with each pair being orthogonal to one another (Kumar *et al.,* 2018; Shi *et al.,* 2018). However, the disadvantage of this approach is that the parallel lines may not be able to function as expected in bad weather conditions such as a blurry vision caused by fog. Another con of this approach is that it takes a lot of manual effort in labelling road surfaces with parallel lines that is needed to correct the perspective distortion effects.

C. Challenges in Perspective Distortion

In Deng *et al.* 2018, in their research work titled "Detection and Recognition of Traffic Planar Objects Using Colorized Laser Scans and Perspective Distortion Rectification", proposes a reliable detection and recognition method for planar objects through the combination of colorized laser scan, color space aggregation (CSA), which consists of RGB, HSV, and CIE L*a*b* color spaces, and perspective distortion rectification. This was further broken down into the following steps; First, colorized laser scans are acquired through data association of both LIDAR and camera. LIDAR, which stands for "Light Detection and Ranging" or "Laser Imaging, Detection and Ranging" is a method for determining ranges by targeting an object or a surface with a laser and measuring the time for the reflected light to return to the receiver. Second, planar objects including traffic signs, street signs, and road surface are detected directly in 3D natural scenes on the basis of aggregation-based feature vector, which concatenates CSA and laser reflectivity (LR) from colorized laser scans. Third, for each of object descriptions or bounding boxes, the possible perspective deformations were corrected to produce front-parallel view with fixed size. While the experimental results of the approaches employed in the research work of these authors performed well under a variety of challenging

conditions, it does not do too well when it comes to accurately detecting overlapping objects in the scene or frame, thus, there is room for further research in developing an algorithm that will better control the challenges posed by perspective deformations.

D. Geometric Based Approaches

In Fatima *et al.,* (2019), in their research work titled "Vehicle Speed Estimation Using Image Processing" aimed at determining vehicle speed which is necessary for traffic surveillance systems. These systems are very much useful to monitor and manage various traffic conditions such as traffic management, prevention of accident, also secure transportation. The paper describes a comprehensive approach to localizing target vehicles in video under various circumstances. The extracted geometry features from the video are continuously projected onto a profile and are constantly tracked. They relied on temporal information of features and their motion behaviors for vehicle identification, which compensates for the complexity in recognizing vehicle shapes, colors, and types. To overcome the disadvantages of traditional methods of vehicle speed estimation, they used image processing done with Python language and OpenCV library. In their paper, the proposed method consists of few basic steps — background subtractions, feature extraction, vehicle tracking etc. The speed is determined using distance travelled by vehicle over number of frames and frame rate. The authors believed that intrusive sensors which are usually based on loop detectors - although widely used in the past, have complex installation and high maintenance cost, promotes asphalt deterioration and can also be damaged by wear and tear. It also requires a direct line of sight connection between vehicle and the equipment. Apart from this, due to high cost of equipment and less accuracy, intrusive sensors as a means of vehicle speed estimation are gradually losing its popularity. Non-intrusive sensors, which include laser meters and doppler radars, avoid these limitations of intrusive sensors, however, they are usually more expensive and require frequent maintenance. Hence, the authors aimed to overcome the challenging task of speed determination by providing an economic solution. Due to the availability of cheaper cameras which are able to produce images with higher quality, video-based systems are by far the most inexpensive alternate for non-intrusive speed measurement.

3. METHODOLOGY

The speed estimation model proposed in this paper consists of a simulation tool, data collection, data analysis, object detection, tracking, refinement and, finally, speed estimation. The focus is on ensuring the mitigation of perspective distortion. The architecture of the speed estimation model is shown in Fig. 1.

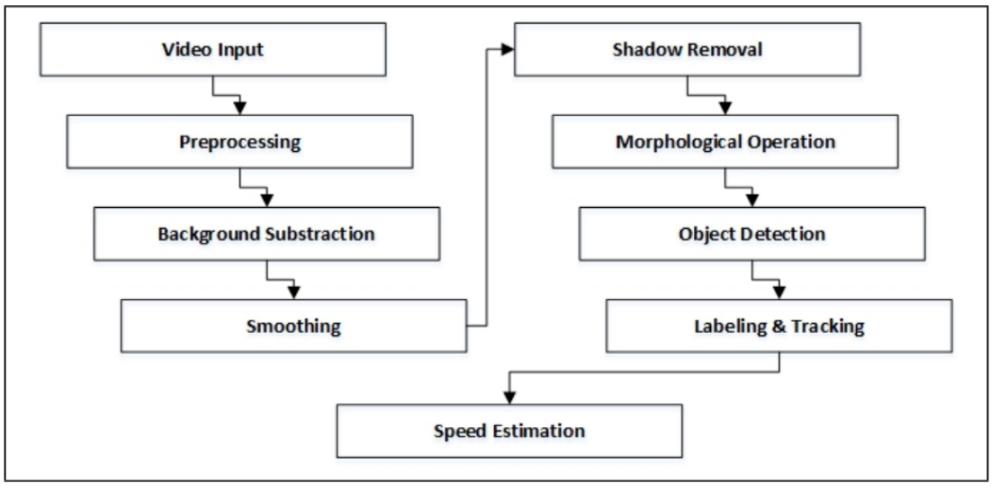


Fig. 1. Architecture of a Vehicle Speed Estimation Model

A. Programming Language and Integrated Development Environment (IDE)

The programming language used in developing the algorithm is Python programming language. It is very useful when working with computer vision techniques that involves object detection, object tracking, and region of interest (ROI) mapping. All these techniques were developed using Python. The integrated development environment (IDE) used is Visual Studio Code, which is a great tool for developing the algorithm using Python programming language and it also provides a simulation environment for the analyzed dataset.

B. Data Collection

The dataset utilized in this study was sourced from supervision's asset utility, which comprises video footage captured from traffic surveillance wide-angle cameras. The dataset available on supervision's asset utility has been captured by stationary wide-angle cameras located at urban intersections and freeways.

The following are details of the dataset:

i. The resolution of the video footage is 1080p (3840x2160) recorded at 30 frames per seconds (fps).
ii. The dataset is 21 seconds long, and the moving vehicles on the highway are trucks and cars.

C. Pre-processing

This phase is an important step in the vehicle speed estimation algorithm; the aim is to enhance the quality of input data and prepare it for subsequent analysis. This step involves a series of transformations and enhancements applied to raw video footage to ensure optimal performance of object detection and tracking algorithms.

D. Data Analysis Methods and Evaluation

This section contains the methods employed to analyze the data collected from supervision's assets utility. These methods are crucial for extracting meaningful insights and evaluating the performance of our algorithm.

i. Object Detection

Object detection in computer vision is the process of identifying and localizing objects within an image or video. It involves detecting the presence of objects in an image and drawing bounding boxes around them to precisely outline their locations.

For the algorithm, object detection is the very first computer vision technique that was applied. It helps to identify where vehicles are located within each frame of the video footage.

Python programming language and YOLO (You Only Look Once) as the object detection model was used. YOLO is known for its efficiency and accuracy in detecting objects in real-time applications. With the YOLO functionality included in the code algorithm, it detects bounding boxes around vehicles in the video footage.

In the algorithm, the bounding boxes was applied using the following syntax;

```
bounding_box_annotator = sv.BoundingBoxAnnotator(thickness=thickness)
```

ii. Object Tracking

Unfortunately, object detection alone is not enough to perform vehicle speed estimation. To determine the distance travelled by each object in the frame, it is important to be able to track them. To achieve that, a functionality that was used is supervision's "ByteTrack" in the algorithm, this functionality assigns a unique tracker ID to each object, enabling the continuous following of the object's motion path across different frames. In the code algorithm, the ByteTrack functionality is represented as;

```
byte_track = sv.ByteTrack(frame_rate=video_info.fps)
detections = byte_track.update_with_detections(detections=detections)
```

iii. Polygon Zone Mapping

This is the next step in the methodology. The observation is that the algorithm creates vehicle detections that are too far away from the camera. These detections tend to be less stable, blinking intermittently, and every time this happens, it risks that the tracker will assign a tracker ID for a vehicle with an unstable detection. To get rid of those unwanted detections, supervision's functionality called "Polygon Zone" filtering feature was used in the code algorithm. The polygon zone is represented as;

```
Polygon_zone = sv.PolygonZone(SOURCE, frame_resolution_wh = video_info.resolution_wh)
```

A polygon zone in computer vision is like drawing a shape on a map to mark a specific area you want to keep an eye on for observation. It helps the algorithm to monitor what is happening inside that area or zone, such as tracking when vehicles enter or exits the zone. This can be seen in Figure 4 below; the actual length of the zone is **250m** and the width across the zone is **25m**. These measurements of the polygon zone are crucial parameters for speed estimation.
The process flow is shown in the Fig. 2.

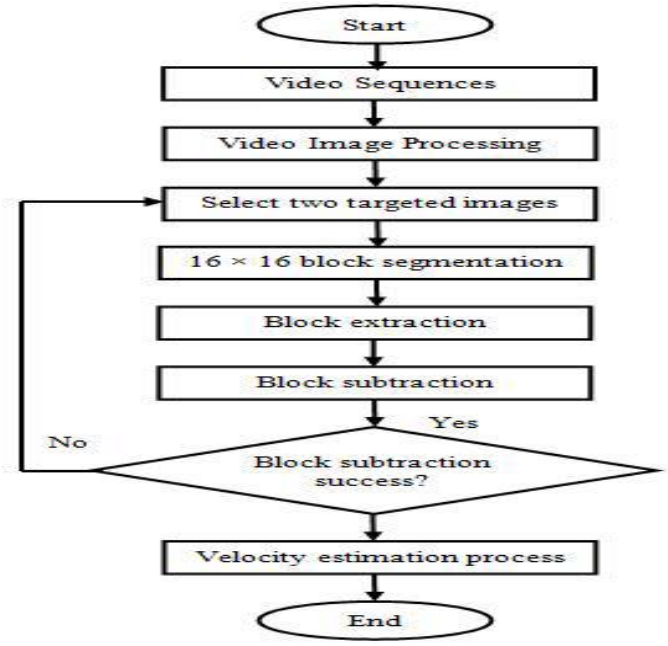


Fig. 2: Flowchart of a Vehicle Speed Estimation Model

4. RESULT AND DISCUSSION

In this chapter we describe the evaluation result of vehicle speed estimation model. First, we introduce the resource needed in experiment, include hardware, software, and OS environment, and then narrate the experiment field and process, finally demonstrate and analyze the experimental result.

I. Introduction of Experimentation Tools

To effectively execute the vehicle speed estimation model, a Windows-based machine with the following hardware and software specifications is recommended:

- CPU: *Intel(R) Xeon(R) CPU E5-2630 v4 @ 2.20GHz*
- Memory (RAM): *Minimum 16 GB (32 GB recommended for large datasets)*
- Software Environment: *Python 3.8, OpenCV, Ultralytics YOLOv8, Supervision (BYTETrack), NumPy, Matplotlib, and other supporting libraries*

II. Object Detection for Speed Estimation

For this implementation, we used the `yolov8x-640` model from Roboflow's pre-trained suite which handles video tasks such as frame extraction, annotation, and result visualization. YOLOv8 is a state-of-the-art object detection model that balances speed and accuracy, making it ideal for real-time applications. The yolov8x-640 variant, with its 640x640 input size, is particularly effective at detecting small and distant objects. The model creates bounding boxes on the detected objects which in this case are vehicles. This can be seen in the Fig. 3.



Fig. 3. Bounding Boxes on Detected Objects in the Frame

III. Multi Object Tracking for Speed Estimation

Object detection alone is insufficient for calculating vehicle speed, as it only identifies vehicles in individual frames. To estimate speed, we need to track each vehicle across multiple frames to measure the distance travelled. For this purpose, we utilize BYTETrack, a robust multi-object tracking algorithm available through the Supervision library's pip package. BYTETrack is designed to maintain consistent tracking of vehicles, even in challenging conditions like occlusions and varying lighting. By tracking vehicles frame by frame, BYTETrack enables precise calculation of their movement over time, which is essential for accurate speed estimation.

IV. Logic Behind Perspective Transformation

To compensate for perspective distortion, we use a transformation matrix, which is computed using OpenCV's getPerspectiveTransform function. This function requires two sets of corresponding points: one from the source region of interest and another from the target region. In our implementation, these regions are denoted as points A-B-C-D and A'-B'-C'-D', respectively. For perspective correction, we select a specific stretch of road from a single video frame as the source region of interest. Highways often have vertical posts or markers spaced at fixed intervals, in this case, every 50 meters. Our region of interest encompasses the full width of the road and the section connecting six of these posts. Given that we are working with an expressway, measurements from Google Maps indicate that the area within the source region is approximately 25 meters wide and 250 meters long. This information allows us to precisely define the vertices of the corresponding quadrilateral, anchoring our new coordinate system at the upper left corner. Finally, we organize the coordinates of the vertices A-B-C-D and A'-B'-C'-D' into 2D matrices representing the SOURCE and TARGET regions. Each row in these

matrices corresponds to the coordinates of a specific point, enabling the calculation of the transformation matrix for accurate perspective correction.

Mathematical Representation of Perspective Transformation

1. Define Points in Source and Target Regions:

Let the four corner points of the **source quadrilateral (in image coordinates)** be:

$$\text{SOURCE} = \begin{bmatrix} xA & yA \\ xB & yB \\ xC & yC \\ xD & yD \end{bmatrix}$$

Let the corresponding four corner points in the **target plane (real-world coordinates)** be:

$$\text{TARGET} = \begin{bmatrix} \text{xA}' & \text{yA}' \\ \text{xB}' & \text{yB}' \\ \text{xC}' & \text{yC}' \\ \text{xD}' & \text{yD}' \end{bmatrix}$$

2. Compute the Perspective Transformation Matrix:

Using OpenCV's getPerspectiveTransform or mathematically solving for the homography matrix **H**, we compute:

$$\boldsymbol{H} = getPerspectiveTransform(SOURCE, TARGET)$$

Here, **H** is a 3×3 matrix such that for any point (x,y) ∈ SOURCE, its corresponding point in the TARGET plane is given by:

$$\begin{bmatrix} x' \\ y' \\ w \end{bmatrix} = \text{H} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad \text{and} \quad (\text{x}', \text{y}') = \left(\frac{x'}{w}, \frac{y'}{w} \right)$$

3. Define Real-World Dimensions

Given:

- Width of the road = 25 meters
- Length of the stretch = 250 meters

Assume A' is anchored at the origin (0,0), then:

$$\text{TARGET} = \begin{bmatrix} 0 & 0 \\ 25 & 0 \\ 25 & 250 \\ 0 & 250 \end{bmatrix}$$

This means the transformed region will appear as a **rectangle** in the real-world coordinate system, eliminating perspective distortion.

V. Calculating Speed

We now have our detector, tracker, and perspective conversion logic in place. It's time for speed calculation. In principle, it's simple: divide the distance covered by the time it took to cover that distance. However, this task has its complexities. In one scenario, we could calculate our speed every frame: calculate the distance travelled between two video frames and divide it by the inverse of our FPS (frames per second), in our case, 1/25. Unfortunately, this method can result in very unstable and unrealistic speed values. To prevent this, we average the values obtained throughout one second. This way, the distance covered by the car is significantly larger than the small box movement caused by flickering, and our speed measurements are closer to the truth.

Mathematical Representation of Speed Calculation

1. Basic Speed Formula

The speed ***v*** is given by:

$$v = \frac{d}{t}$$

Where:

- ***d*** is the distance traveled (in meters)
- ***t*** is the time taken (in seconds)

2. Instantaneous Speed per Frame (Unstable Version)

Assume:

- $P_t = (x_t, y_t)$ is the position of the vehicle at frame ***t***
- $P_{t+1} = (x_{t+1}, y_{t+1})$ is the position at the next frame
- Δd_t is the real-world distance between these positions (after perspective transformation)
- FPS = 25 ⇒ $\Delta t = \frac{1}{25}$ sec

Then:

$$V_t = \frac{\Delta dt}{\Delta t} = \Delta d_t \cdot \text{FPS}$$

But this yields **noisy speed values**, as Δd_t may fluctuate due to small detection shifts. To stabilize measurements, average the speed over ***N*** frames (e.g., 25 frames for 1 second at 25 FPS).

Let:

- $\Delta D = \sum_{i=t}^{t+N-1} \Delta d_i$, , total distance covered over N frames
- $\Delta T = \frac{N}{FPS}$ = 1 second

Then:

$$\bar{v} = \frac{\Delta D}{\Delta T} = \sum_{i=t}^{t+N-1} \Delta d_i \cdot \frac{FPS}{N}$$

Since $\Delta T = 1$, the formula simplifies to: $\bar{v} = \Delta D$

So, the speed after smoothing becomes the **total distance covered in one second** — much more stable and realistic.

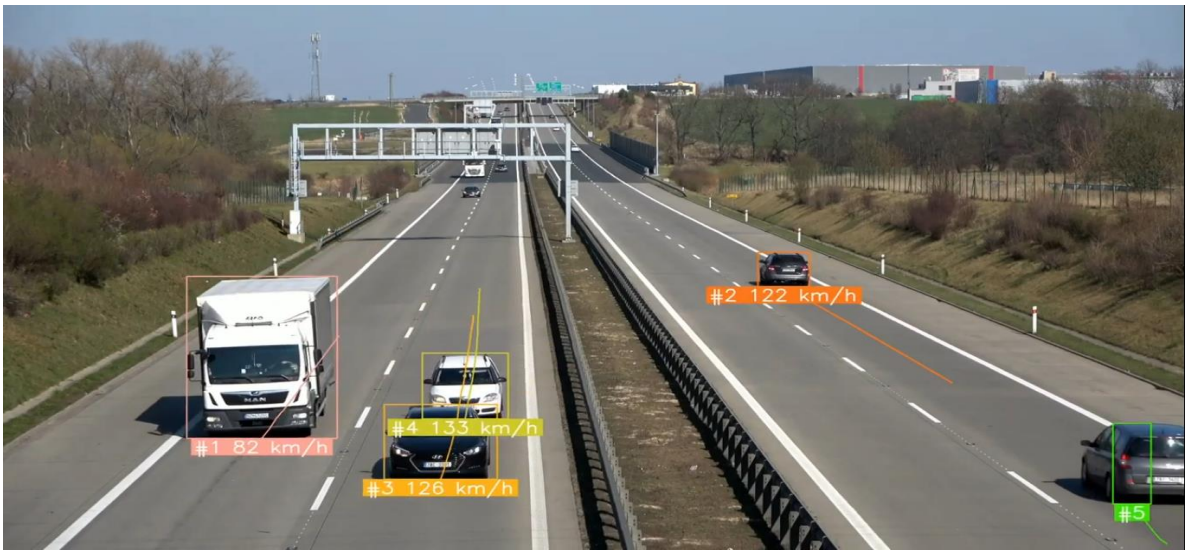


Fig. 4. Output Image of Speed Detection

VI. Evaluation of Methodology

Performance Metrics Summary

The following table summarizes the performance of the vehicle speed estimation model across the key metrics:

Metrics	Accuracy	Precision	Recall	F1 Score
YOLO Model	0.80	0.90	0.82	0.86

Tab 1: Model Performance Metrics

Visual Representation of Metrics

To further illustrate the model's performance, the following bar chart provides a visual comparison of the four evaluated metrics:

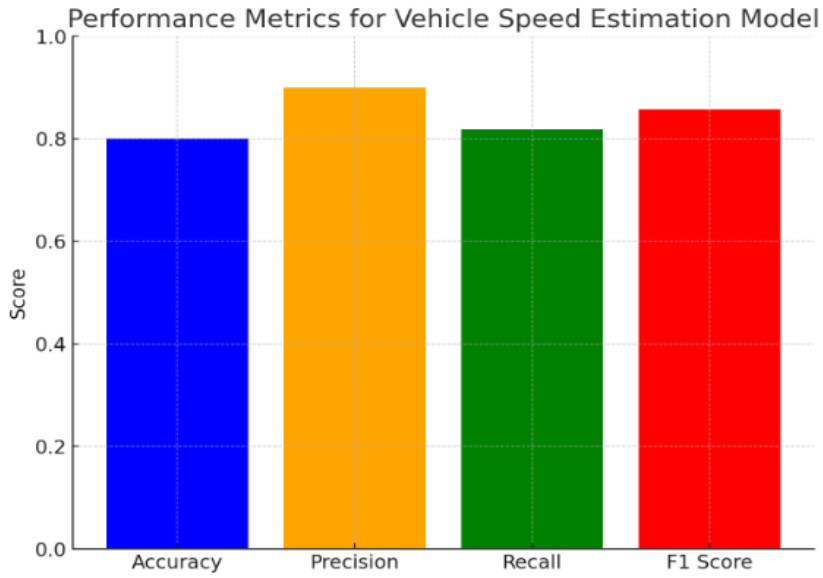


Fig. 5. Bar Chart Indicating the Performance Metrics

5. CONCLUSION

A key objective was to address the challenge of perspective distortion, which can significantly affect the accuracy of speed measurements in video footage. The implementation began with the use of the YOLOv8 model for object detection. YOLOv8, with its high-resolution detection capabilities, was employed to identify and locate vehicles within each frame of the video. This step was crucial for establishing the initial positions of vehicles. To correct perspective distortion and accurately estimate vehicle speed, a transformation matrix was applied. This matrix was computed using OpenCV's `getPerspectiveTransform` function, which aligned the source region of interest (defined by specific markers on the road) with the target region. This correction enabled a more accurate mapping of vehicle movements within the video frames. For tracking vehicles and calculating the distance traveled, the BYTETrack algorithm was integrated into the model. BYTETrack efficiently maintained vehicle identities across frames, even under challenging conditions, allowing for precise measurement of displacement and speed. The performance of the implemented system was evaluated based on key metrics such as accuracy, precision, recall, and F1 score. The results demonstrated significant improvements in speed estimation accuracy, validating the effectiveness of addressing perspective distortion through the proposed approach.

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